

Nicholas C. Lopes

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EXPERIENCE

Embry-Riddle Aeronautical University	Daytona Beach, FL
<i>Graduate Research Assistant — Thermal Science Laboratory</i>	Jan 2021 – May 2026
• Performed CFD of supercritical flows & latent storage processes to evaluate relevant thermal behavior • Mentored 10+ undergraduate and graduate researchers and established best practices for the team • Synthesized results into 20+ publications and presented findings at technical conferences • Wrote research proposals and secured \$10,000+ in project funding	
Oak Ridge National Laboratory (ORNL)	Oak Ridge, TN
<i>Thermal Storage Research Intern — Multifunctional Equipment Integration Group</i>	Jan 2025 – Dec 2025
• Led end-to-end CFD modeling of a DOE-funded heat pump water heater with embedded phase change capsules • Developed simulation-ready CAD geometry from experimental hardware and established solver workflows • Coordinated with ORNL researchers to translate experimental data and specifications into validated CFD models • Performed sensitivity and parametric studies, improving phase change capsule utilization by 50+%	
German Aerospace Center (DLR)	Cologne, Germany
<i>Combustion Research Intern — Institute of Propulsion Technology</i>	Jun 2024 – Aug 2024
• Designed and optimized a novel cooling hole array for a fuel-lean combustor liner • Built a full-cycle CFD workflow using DLR's high-performance computing systems • Collaborated with DLR experimentalists to benchmark and validate simulations against combustor test data • Improved cooling performance by 15+% and published findings in peer-reviewed journal and conference proceedings	
Pratt & Whitney	West Palm Beach, FL
<i>Aero-Thermal Engineer Intern — Aero-Thermal Fluids Group</i>	Jun 2022 – Aug 2022
• Conducted CFD/FEA of PW4000 and F-135 hot-section frames and cases • Integrated multiple CAD models into a unified combustor-turbine assembly and defeatured geometry for analysis • Interfaced with structures team and reported thermal profiles for life-cycle assessments • Developed code to automate data processing and organization for auxiliary power unit systems	

PROJECTS

A3 Refrigerant Leakage Analysis	Jan 2026 - Jun 2026
<i>Oak Ridge National Laboratory (ORNL)</i>	
• Evaluated A3 refrigerant leakage behavior from critical HVAC components with high-fidelity CFD • Delivered key insights to ORNL including gas dispersion and ignition likelihood to enhance mitigation strategies	

EDUCATION

Embry-Riddle Aeronautical University	Daytona Beach, FL
<i>Doctor of Philosophy in Mechanical Engineering</i>	Aug 2023 – May 2026
<i>Master of Science in Mechanical Engineering</i>	Jan 2021 – May 2023
<i>Bachelor of Science in Aerospace Engineering</i>	Aug 2017 – Dec 2020

SKILLS

Programming & Version Control: MATLAB, Python, C, Scheme, Perl, Git
CAD Software: SolidWorks, NX, ANSYS Spaceclaim, ANSYS Discovery, ANSYS Design Modeler
CFD & FEA Tools: ANSYS Fluent, Pointwise, ANSYS Mechanical, ANSYS Workbench, ANSYS CFX, Tecplot
High-Performance Computing: Linux environments, Torque & Slurm job schedulers, Bash/shell scripting